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Feature space expansion and compression with spatial–spectral augmentation for hyperspectral image Class-Incremental Learning

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ABSTRACT

Hyperspectral image classification (HSIC) has gained significant attention because of its applications in geographic detection and military surveillance. Numerous deep neural network-based approaches have been proposed to enable efficient HSIC. However, changing landscapes and sequentially acquired scenes require HSIC methods that can continuously learn new classes. Updating models with new data often leads to catastrophic forgetting of previously learned knowledge. Existing methods struggle to maintain consistent classification performance across all classes during the progressive updating process. To address this challenge, we propose a learning framework named Feature space Expansion, Integration, and Compression with spatial–spectral augmentation for Class-Incremental Learning (FEICA-CIL). This framework aims to enable the retention of old knowledge and generalization of new knowledge simultaneously. The spatial–spectral mixed augmentation technique encourages the model to explore robust representations in two dimensions. The CIL training process is divided into two stages: initial and incremental. An online adverse distillation strategy is introduced to optimize the capacity of the model in the initial stage. In the incremental stage, an expansion and integration strategy is proposed to enlarge the feature space for new tasks while retaining old knowledge. A compression approach is employed to simplify the expanded structures. The incremental stage can be repeated to increase the capability of the model continuously. Extensive experiments were conducted on four datasets. Our method exhibited novel performance compared with other IL frameworks on hyperspectral data, surpassing comparative approaches, with the largest improvement being 1.65%. In addition, ablation studies demonstrated the effectiveness of the proposed modules. The code is available at https://github.com/knockshot/FEICA-CIL.

1. Introduction

Hyperspectral image classification (HSIC) is an important topic in pattern recognition and has been widely studied in several fields, including agricultural research, ocean exploration, military surveillance, and mineral detection. The ultimate aim of HSIC is to assign every pixel in the hyperspectral image to a specific category based on its spatial and spectral characteristics; thus, HSIC is a pixel-level image classification task. Traditional machine learning methods, such as support vector machines [1] and multinomial logistic regression [2], were initially adopted in HSIC. These approaches focus on utilizing the spectral features of hyperspectral data. Dimension-reduction methods have also been proposed for extracting more representative features from spectral channels, including principal component analysis (PCA) [3] and independent component analysis [4]. Spatial–spectral classification methodologies have been introduced to realize the synchronous utilization of spatial and spectral signatures. MSMIL [5] introduced a spatial–spectral decision fusion strategy to weigh multi-scale superpixel maps. Functional data analysis has also been incorporated into HSIC to extract better spatial and spectral knowledge [6].

Inspired by the dominance of deep learning in other optical computer vision tasks, many HSIC approaches based on deep neural networks have been developed to improve the classification performance in hyperspectral data tasks. Nevertheless, they were based on the premise that the training dataset could cover all types of targets in testing scenarios. However, the landscape is usually affected by human activities and seasonal changes, and it changes gradually over time. Besides, different landscape types may be obtained sequentially during training. Hence, the models must continuously increase their capacity to accommodate dynamic landscape changes. However, the catastrophic forgetting phenomenon [7] is unavoidable if models are updated with new incoming data. The previous decision boundary will change if the model is generalized to new image types.

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Class-incremental learning (CIL) is proposed to address catastrophic forgetting by enabling models to continuously learn new classes while retaining knowledge of previously learned ones. As the most challenging scenario in incremental learning, CIL focuses on developing a unified classifier that can effectively discriminate among all encountered classes, regardless of when they were introduced to the model. CIL approaches can be categorized into parameter regularization, structural, rehearsal, and knowledge distillation methods. Prior works in this field include RWalk [8], which iteratively updates the fisher information matrix using moving average techniques to develop an approach that maintains stability regardless of task quantity; KAN [9], which expands the feature extractor during the learning process and introduces an adaptive feature fusion module to dynamically balance and integrate information from multiple branches; Tang et al. [10] developed a learnable feature generator to create diverse exemplars by combining semantic information from exemplars with semantically-irrelevant information from unlabeled data; and K3D [11], which employs knowledge fusion distillation to maintain clear decision boundaries and reduce feature confusion.

While these approaches have advanced CIL in general domains, they are not specifically designed to address the unique challenges of HSIC, which involves high-dimensional spectral data with complex spatial–spectral relationships. To overcome catastrophic forgetting in CIL specifically for HSIC applications, we propose Feature space Expansion, Integration, and Compression with spatial–spectral Augmentation (FEICA-CIL). The framework first introduces a spatial–spectral mixed augmentation technique that enhances the model robustness by extracting comprehensive representations from both the spatial and spectral dimensions, directly mitigating catastrophic forgetting through improved cross-class performance. The learning process is structured into two distinct stages, namely initial and incremental, with each stage optimized through network quantization to reduce the computational overhead. During the initial stage, we implement an online adverse distillation mechanism where a student network learns simultaneously from ground truth labels and a teacher network, while the teacher explores knowledge beyond the scope of the student. This distillation strategy enables more abstract representations, allowing the model to learn robust knowledge and maintain strong performance on the first task, thereby mitigating catastrophic forgetting. In the incremental stage, we address feature space limitations by introducing a new full-precision model that expands feature space. This expansion allows clear decision boundaries for new classes while preserving old class representations through a model ensemble. The framework alleviates catastrophic forgetting by supplementing, rather than modifying, the feature extractor for previously learned data. Finally, we compress the dual-model ensemble into an efficient single-branch network through knowledge distillation, optimizing both the performance and computational efficiency. The incremental stage can be repeated for further enlargement of the model's capacity. The main contributions of this study are summarized as follows:

- This study proposes a CIL framework named FEICA-CIL, which incrementally trains a model for HSIC. In particular, a spatial–spectral mixed augmentation technique is introduced to enable the model to extract more robust features. The training process is divided into two stages according to the optimization objectives. Different optimization strategies are applied for different stages.
- In FEICA-CIL, a full-precision network is introduced as a teacher model to provide online supervision for training the lightweight network in the initial stage. In the incremental step, a new full-precision network is introduced to narrow the gap between the old and target feature spaces. By combining the old and new models, the features of the previous classes are maintained, while the features of the new categories are well formulated. The ensemble of the two models is distilled into a single-branch model to reduce the growth of structures.
- Extensive experiments were conducted on four datasets. The results demonstrate the effectiveness of the proposed method. Specifically, our approach exceeds all other comparative objects in all training settings, even when we migrated the same backbone to other CIL methods. The largest margin was achieved on the Hanchuan dataset, with an improvement of 1.65%.

The remainder of this paper is organized as follows: Section 2 reviews HSIC mechanisms and recent CIL research. Section 3 presents the proposed FEICA-CIL framework. Section 4 details experimental setup and results on multiple hyperspectral datasets. Section 5 concludes the paper and discusses future work.

2. Related works

In this section, related studies on HSIC and CIL are elaborated.

2.1. Hyperspectral image classification

Deep learning has revolutionized HSIC by extracting hierarchical features, from low-level textures and edges in earlier layers to complex abstract features in deeper layers [12]. Common architectures include stacked autoencoders (SAEs), deep belief networks (DBNs), recurrent neural networks (RNNs), and convolutional neural networks (CNNs).

The SAE consists of multiple concatenated autoencoders (AEs). The classification results can be obtained using a subsequent logistic regression classifier. LAutoAE [3] is an efficient lightweight AE that addresses computational challenges. DBN employs the Restricted Boltzmann Machine (RBM) as its fundamental building block, providing an efficient solution to mitigate the challenges of slow convergence and overfitting that are prevalent in conventional neural network architectures. CG-based DBN [13] adopts the gradient descent algorithm to accelerate the DBN convergence and avoid the “zig-zagging” problem. The RNN can capture the dynamic temporal characteristics of sequential data and perform classification through a recurrent hidden state. Geo-DRNN [14] combines a U-Net with an RNN to mimic the human brain and continuously refines the output classification map.

The CNN is the most widely used deep learning method for HSIC. CNNs can extract more robust representations through deep hierarchical structures to realize better classification performance. AAtt-CNN [15] achieves the automatic design and optimization of CNNs through a neural architecture search and channel-based attention mechanisms. GMHN [16] integrates global and local information and utilizes superpixel features in HSIC. Spatial information was integrated into HSIC in [17]. The proposed model incorporates both spatial and spectral classifiers.

2.2. Class-incremental learning

Most CIL approaches can be categorized into four types according to their anti-forgetting techniques: parameter regularization, structural, rehearsal, and knowledge distillation methods.

Parameter regularization methods condition the updating of weights using regularization constraints. EWC [18] calculates the Fisher information matrix to estimate the importance factors for each weight. The renewal of the model in the current task is regularized by the weights from the previous model with the importance factors as coefficients. Structural approaches explore the network architecture to realize old knowledge reservations. Some methods, such as HAT [19], use gated parameters to maintain significant parameters for previous tasks. Besides, some approaches, such as DNE [20], expand the model to integrate new knowledge. Rehearsal methods concentrate on selecting representative samples to reproduce the domain distribution of the entire training dataset. For example, DER [21] incorporates a random selection strategy to ensure equal probability for each sample to be selected. Knowledge distillation approaches follow the basic knowledge distillation methodology. A model that is well-trained on previous tasks is considered as the teacher model to guide the optimization in the current task. PODNet [22] utilizes pooled intermediate features to realize relaxed distillation constraints. FOSTER [23] introduces new modules to narrow the residuals between targets and outputs and to compress the inference block further using distillation strategies.

CIL has been studied increasingly in HSIC in recent years. LPILC [24] introduces a linear programming incremental learning classifier to adapt previous models to new datasets. Meng et al. [25] combined knowledge distillation and linear correction to balance the bias over the old and new classes in the outputs of incremental models. GS_2FIN-CIL [26] consists of a new network architecture based on an attention block to get spatiospectral features. A bias correction mechanism is adopted along with the BCK classifier to correct the bias towards new classes. These studies demonstrated the potential of CIL in continuously learning knowledge from hyperspectral data and obtaining superior classification results.

Traditional class-incremental learning (CIL) enables models to expand their classification capacity when sufficient new samples become available. In scenarios where exemplars from new classes are limited, few-shot class-incremental learning (FSCIL) provides a solution for acquiring new knowledge. The primary challenges of few-shot incremental learning include model overfitting on new data and catastrophic forgetting of previously learned information. TOPIC [27] addressed these challenges by leveraging a neural gas network to preserve existing knowledge while strengthening the representation learning of new categories. CPC [28] extended FSCIL to remote sensing imagery by introducing a prototype separability module that enhances inter-class distinguishability and generates more representative prototypes for new data. In summary, while both CIL and FSCIL address incremental learning challenges, they do so under different data availability assumptions. Our proposed method operates within the CIL framework, where adequate new samples are available for training.

3. Methodology

The details of the proposed FEICA-CIL method are fully discussed in this section. Specifically in the section. 3.1, the entire architecture of FEICA-CIL is illustrated and the training process is demonstrated. In addition, the novelties of FEICA-CIL are introduced. The augmentation technique assists the model in capturing more robust characteristics from both the spatial and spectral dimensions. The distillation strategy used in the initial stage is explained, and the expansion and compression processes in the incremental step are elaborated upon. Table 1 presents the symbols used in this section, along with explanations of their functions.

3.1. Framework

Fig. 1 depicts the framework of the proposed FEICA-CIL. The process begins with extracting small data cubes from hyperspectral images using a sliding window, followed by applying spatial–spectral mixed augmentation to these selected samples. The framework operates in two stages: In the initial stage, a quantized student model is guided by a full-precision teacher network. During the incremental stage, when new classes are introduced into the training dataset, the previously trained model is preserved and combined with a new full-precision network to form an ensemble. This ensemble integrates the feature extraction capabilities of both models. To address the increasing complexity of the parameters, a new lightweight model is trained from scratch to replicate the inductive characteristics of the ensemble. This incremental stage can be repeated iteratively based on the requirements for expanding the training samples. Besides, the hyper-feature aggregation loss functions [29] and local similarity classifier [22] are adopted for the models in all stages. The feature aggregation loss function is formulated as follows:

Table 1
Symbols used in this section and their corresponding explanations. The difference between the feature extraction backbone and backbone network is whether feature promotion layers are considered. The feature extraction backbone generates more abstract and representative features aided by the feature promotion layers.

Symbol	Explanation
$\langle\cdot\rangle$	cosine similarity function
$[\![\cdot]\!]$	blocked gradient path
$[\cdot]_+$	activation function
MSE$[\cdot]$	Mean Square Error loss function
θ	the feature extraction backbone in the initial stage
Θ	the backbone network in the initial stage
$\mathcal{G}$	the feature promotion layer
$\mathcal{S}$	negative cosine similarity operator
x and $\bar{x}$	augmented samples
y	the label of the sample
τ	cosine similarity operator
μ	the parameter conditioning the new model's output in the expansion stage
η	the vector conditioning the old model's output in the expansion stage
Θ_o	the backbone of the old model in the expansion stage
Θ_n	the backbone of the new model in the expansion stage
Θ_{en}	the ensemble of Θ_o and Θ_n
Φ	overall feature extraction backbone including Θ_{en} in the expansion stage
ϑ	the feature extraction backbone of the new model in the compression stage
Θ_q	the backbone of the new model in the compression stage

$$\mathcal{L}_{ha}(\theta) = \frac{1}{2k}\sum_i\sum_j \mathbb{1}\left(y_i == y_j\right)\cdot\mathcal{S}\left(\theta(x_i),\theta(\bar{x}_j)\right) \tag{1}$$

$$\begin{aligned}\mathcal{S}\left(\theta(x_i),\theta(\bar{x}_j)\right) = &\left(1-\left\langle [\![\mathcal{G}_1\left(\Theta(x_i)\right)]\!],\mathcal{G}_2\left(\mathcal{G}_1\left(\Theta(\bar{x}_j)\right)\right)\right\rangle\right)\\ &+\left(1-\left\langle [\![\mathcal{G}_1\left(\Theta(\bar{x}_j)\right)]\!],\mathcal{G}_2\left(\mathcal{G}_1\left(\Theta(x_i)\right)\right)\right\rangle\right)\end{aligned} \tag{2}$$

The indicator function, denoted as $\mathbb{1}$, returns a value of 1 when its input satisfies the specified requirement. θ is the feature extraction backbone of the model, including the backbone network and feature promotion layers. Θ represents the backbone network and $\mathcal{G}_1$ and $\mathcal{G}_2$ are two feature promotion layers. $\mathcal{S}$ denotes a negative cosine similarity operator based on the representations after the feature promotion layers. The feature promotion layers project the features onto higher dimensions and identify more typical representations. $\langle\cdot\rangle$ denotes the cosine similarity function. $[\![\cdot]\!]$ implies that the gradient path of the inner part is blocked. x and $\bar{x}$ represent the augmented samples, and y is the sample label. k is a variable calculated from accumulated squared lengths of each class. This function aggregates the representations of samples from the same class in the hyper-feature space. These features do not implicitly converge. The diversity of the embeddings is retained, which is beneficial for training the classifier. The local similarity classifier is formulated as follows:

$$s_{c,k} = \frac{\exp\left\langle \boldsymbol{v}_{c,k},\mathbf{h}\right\rangle}{\sum_i \exp\left\langle \boldsymbol{v}_{c,i},\mathbf{h}\right\rangle},\quad \hat{\mathbf{y}}_c = \sum_k s_{c,k}\left\langle \boldsymbol{v}_{c,k},\mathbf{h}\right\rangle \tag{3}$$

$\boldsymbol{v}_{c,k}$ represents the kth normalized class agent for the cth class. The model output is denoted as $\mathbf{h}$, and $\hat{\mathbf{y}}_c$ represents the score of $\mathbf{h}$ for the cth class. The outputs of the local similarity classifier can be formulated as follows:

$$\mathcal{L}_{LSC} = \left[-\log\frac{\exp\left(\sigma\left(\hat{\mathbf{y}}_y-\delta\right)\right)}{\sum_{i\neq y}\exp\left(\sigma\hat{\mathbf{y}}_i\right)}\right]_+ \tag{4}$$

The parameter σ is a learnable scaling factor, and δ is a constant used to induce a stronger inter-class separation. The activation function is denoted by $[\cdot]_+$. The basic loss module in the proposed method is as follows:

$$\mathcal{L}_b(\theta) = \mathcal{L}_{LSC}(\theta) + \mathcal{L}_{ha}(\theta) \tag{5}$$

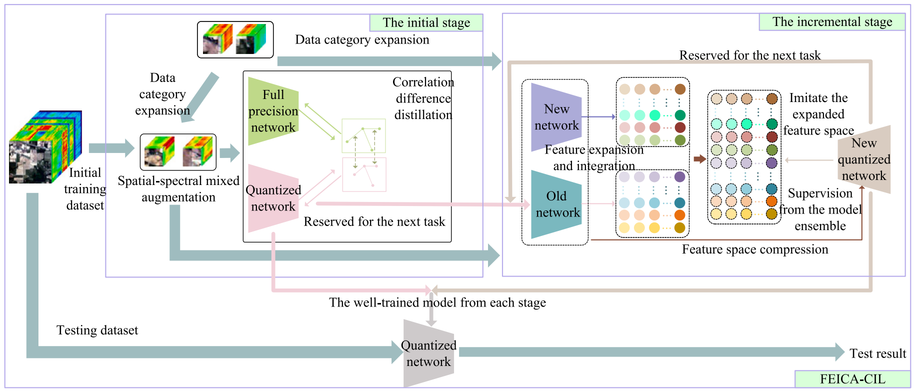

Fig. 1. Illustration of learning phases in FEICA-CIL. Hyperspectral data is divided into small cubes to match the requirements of the model's input. Spatial–spectral mixed augmentation is introduced to improve the model's representation extraction ability. The initial stage is designed for the initial state of the CIL when no old knowledge must be reserved. The incremental stage enables the model to enlarge its classification capacity for new classes while remembering knowledge from old classes.

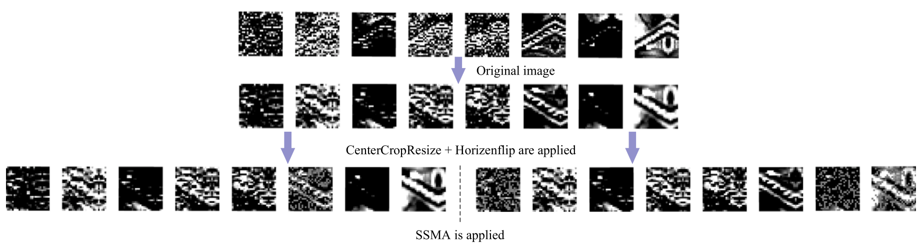

Fig. 2. Visualization of image augmentation process. The upper row shows images of different channels from a single sample. The middle row demonstrates the effect of the *CenterCropResize* and the *Horizonflip*. The bottom row exhibits the impact of SSMA. The augmentations of the same sample have diverse patterns after SSMA. Aggregating these exemplars induces the model to find robust representations from both spatial and spectral dimensions.

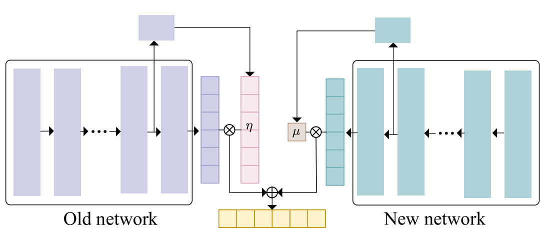

Fig. 3. Process of computing trainable η and μ using internal features. Features from the old model determine the value of η, conditioning the impact of the old model. Representations from the new model control the effect of the new model with μ.

3.1.1. Spatial–spectral mixed augmentation

We propose spatial–spectral mixed augmentation (SSMA), which performs random permutations across both spatial and spectral dimensions by altering values at selected points within randomly chosen bands. Combined with hyper-feature aggregation loss, SSMA encourages learning of stable, representative features. We also incorporate *CenterCropResize* and *Horizonflip* techniques. As shown in Fig. 2, SSMA changes the randomly selected channels in different augmentations, forcing the model to extract robust representations from both the spatial and spectral dimensions.

3.1.2. The initial stage optimization

In the initial stage, the model is trained to undertake a common classification task. Hence, a distillation strategy is employed to improve the classification performance. The common distillation strategy only transfers knowledge from the teacher to the student, requiring a well updated teacher model. Inspired by adversarial distillation, we propose constructing a one-stage framework where the teacher and student models can be updated simultaneously so that our method can be easily migrated to diverse hyperspectral scenarios. Through the adversarial relationship established between their outputs, the teacher model is compelled to emphasize differences from the student model during training, enabling it to explore knowledge beyond the student's current understanding and provide meaningful supervision. This knowledge is then transferred to the student model through conventional distillation constraints. In this way, the teacher model can effectively guide the student's learning without requiring pre-training. We use the correlation relation between samples to capture the feature space of the model. Specifically, assuming that $\mathcal{D} = \{(x_1, y), (x_2, y), (x_3, y), \dots, (x_n, y)\}$ is a dataset including all samples belonging to category y, the outputs of the student model can be organized as:

$$\theta_s(\mathcal{D}) = \left[\theta_s(x_1), \theta_s(x_2), \theta_s(x_3), \dots, \theta_s(x_n)\right] \tag{6}$$

Here, θ_s represents the feature extraction backbone of the student model. Then, the correlation relations of this specific class can be calculated:

$$\mathcal{M}_s = \left\{\tau\left[\theta_{s1}, \theta_{s2}\right], \tau\left[\theta_{s1}, \theta_{s3}\right], \dots, \tau\left[\theta_{sn}, \theta_{s(n-1)}\right]\right\} \tag{7}$$

τ is the cosine similarity operator. θ_{s1} denotes $\theta_s(x_1)$. Similarly, the correlation relations generated by the teacher model are as follows:

$$\mathcal{M}_t = \left\{\tau\left[\theta_{t1}, \theta_{t2}\right], \tau\left[\theta_{t1}, \theta_{t3}\right], \dots, \tau\left[\theta_{tn}, \theta_{t(n-1)}\right]\right\} \tag{8}$$

Let θ_t represent the feature extraction backbone of the teacher model. The difference between $\mathcal{M}_s$ and $\mathcal{M}_t$ is computed as $P_{st} = |\mathcal{M}_s - \mathcal{M}_t|$. While this calculation initially focuses on specific classes, the actual training process involves computing the intra-class correlation matrices independently for each class to obtain the difference matrices. We preserve the top one-third values in P_{st} as coefficients for the teacher model's hyper-feature aggregation loss function. The modified loss function is formulated as follows:

$$\mathcal{L}_t(\theta_t) = \frac{1}{2k}\sum_i \sum_j \mathbb{1}\left(y_i == y_j\right) \cdot P_{st} \cdot \mathcal{S}\left(\theta_t(x_i), \theta_t(\bar{x}_j)\right) \tag{9}$$

This loss serves as the adversarial distillation loss, highlighting the differences between the teacher and student models. Through this mechanism, the teacher model explores knowledge beyond the current understanding of the student model. Simultaneously, we introduce a distillation loss function to facilitate knowledge transfer from the teacher to the student model:

$$\mathcal{L}_{dist}(\theta_s) = MSE\left[\Theta_t(x), \Theta_s(x)\right] + \mathcal{L}_{hd}(\theta_t, \theta_s) \tag{10}$$

$$\mathcal{L}_{hd}(\theta_t, \theta_s) = \frac{1}{2k}\sum_i \sum_j \mathbb{1}\left(y_i == y_j\right) \cdot \mathcal{S}\left(\theta(x_i), \theta(x_i)\right) \tag{11}$$

Θ denotes the backbone of the network, whereas θ represents the feature extraction backbone of the network where the feature promotion layers are included. MSE indicates that the mean square error loss function is adopted. X includes the augmented samples. $\mathcal{L}_{hd}$ aggregates the representations of the student and teacher models in the higher dimension while MSE constrains the outputs in the lower dimension. The overall loss function in the initial stage is as follows:

$$\mathcal{L}_{int} = \mathcal{L}_b(\theta_t) + \mathcal{L}_b(\theta_s) + \mathcal{L}_t(\theta_t) + \mathcal{L}_{dist}(\theta_s) \tag{12}$$

3.1.3. Feature expansion and integration

Although the old model trained in the last stage can well represent the characteristics of the samples from the old classes. New classes are beyond its sight. Hence, a new model is introduced to supplement the representation capability of the old model. Let (x, y) denote the input and its label, with Θ_o and Θ_n representing the backbones of the old and new models respectively. The combined representation of these backbones is denoted as $\Theta_{en}(x)$. The training process can be represented as:

$$\Theta_{en}(x) = \Theta_o(x) + \underset{\Theta_n}{argmin} E_{(x,y)}[l(y, \varphi(\Theta_o(x) + \Theta_n(x)))] \tag{13}$$

Here, $\varphi(\cdot)$ denotes the normalized logits of the model ensemble, and $l(\cdot)$ represents the loss function. Then, the ideal training result should be:

$$y = \varphi(\Theta_o(x) + \Theta_n(x)) = Softmax(\tau(\boldsymbol{v}_{n+m}, \Theta_o(x) + \Theta_n(x))) \tag{14}$$

Let τ denote the cosine similarity function, and $\boldsymbol{v}_{n+m}$ represent the growing class agents in the local similarity classifier. While the new model reduces the disparity between old and ideal features, simply accumulating outputs from both models constrains the update capability of Eq. (14). This limitation occurs because only Θ_n can be updated during incremental training, while the impact of Θ_o remains static. For new tasks, the model ensemble is enhanced to accommodate additional classes. We implement a trainable factor η to independently regulate each channel of the old model's output. η is implemented as a factor rather than a parameter because the disproportionate volume of new class data compared to old class data can drive a trainable parameter toward trivial values during training. For old classes, feature representations are effectively captured by the old model's outputs. The model ensemble is trained to reconstruct feature patterns using selected samples. Therefore, we introduce a trainable factor μ that modulates the influence of Θ_n. The final optimization objective is formulated as follows:

$$\Theta_n = \underset{\Theta_n}{argmin}[y, Softmax(\tau(\boldsymbol{v}_{n+m}, \mu \cdot \Theta_o(x) + \eta \cdot \Theta_n(x)))] \tag{15}$$

Fig. 3 demonstrates how these two factors are obtained during the training. μ acts as a global attention coefficient for the outputs of the new model, while η is a channel attention parameter for the outputs of the old network.

Table 2
The number of training and testing samples of each dataset.

Dataset	Class	Train	Test	Total
SA	16	1076	53053	54129
PU	9	1277	41499	42776
Longkou	9	1019	203523	204542
Hanchuan	16	1279	256251	257530

Setting overall feature extraction backbone including Θ_{en} as Φ, the overall loss function of the feature expansion and integration step is as follows:

$$\mathcal{L}_{integ} = \mathcal{L}_b(\Phi) = \mathcal{L}_{LSC}(\Phi) + \mathcal{L}_{ha}(\Phi) \tag{16}$$

3.1.4. Feature space compression

While the model ensemble achieves impressive classification results, it introduces significant computational overhead during inference. Since a single model can typically handle more classes, parameter redundancy likely exists. To address this, we propose a compression strategy using knowledge distillation. A new quantized model is trained under the supervision of the previously trained model ensemble, employing multi-level feature constraints similar to the initial stage. The optimization objective is formulated as follows:

$$\mathcal{L}_{simp} = \mathcal{L}_b(\vartheta) + \mathcal{L}_{hd}(\vartheta, \Phi) + \mathcal{L}_{fe}(\vartheta) \tag{17}$$

$$\mathcal{L}_{fe}(\vartheta) = \sum -\left[\left\langle \Theta_q(x), \Theta_{en}(x)\right\rangle + \left\langle \Theta_q(\bar{x}), \Theta_{en}(\bar{x})\right\rangle\right] \tag{18}$$

Here, ϑ represents the feature extraction backbone of the new model, which consists of the feature promotion backbone and the backbone network Θ_q. $\langle\cdot\rangle$ represents the cosine similarity function. Unlike the loss in the initial stage, we choose cosine similarity loss rather than the means square error loss to restrict shallow features output by Θ_q.

4. Experiments

The performance of the proposed FEICA-CIL method is evaluated on several datasets. These results demonstrate the novelty of the proposed method. Ablation studies were conducted to analyze the impacts of different modules in the proposed architecture.

4.1. Experimental settings

This section explains the data settings in the incremental learning framework and the parameter settings for training. The running environment is also described.

4.1.1. Data description

Four classical hyperspectral datasets were adopted in the evaluation experiments: SA (Salinas), PU (University of Pavia), Longkou (WHU-Hi-LongKou) and Hanchuan (WHU-Hi-HanChuan). The SA dataset comprises 512 × 217 pixels with 224 spectral bands, containing 16 distinct classes in its ground truth. The PU dataset contains 610 × 340 pixels with 103 spectral bands, with its land cover categorized into 9 classes. The Longkou dataset comprises 550 × 400 pixels with 270 spectral bands, containing 9 land-cover classes. The Hanchuan dataset contains 1217 × 303 pixels with 274 spectral bands, comprising 16 target classes. Dataset-specific splitting ratios were used based on their respective sizes. Specifically, we selected 2% of the SA dataset, 3% of the PU dataset, and 0.5% of the Longkou [30] and Hanchuan [30] datasets, respectively, as training sets. Table 2 lists the numbers of training and testing samples. The PCA approach was adopted to reduce the spectral dimensions to eight. The entire data block was divided into 3D cubes of size 8 × 27 × 27.

Table 3
Classification results of FEICA-CIL on the SA and Hanchuan datasets (%).

Dataset	SA				Hanchuan			
Class	Task 1	Task 2	Task 3	Task 4	Task 1	Task 2	Task 3	Task 4
1	100.00	100.00	100.00	100.00	99.96	99.53	99.82	99.33
2	99.94	100.00	100.00	100.00	99.84	97.86	98.87	98.13
3	100.00	100.00	100.00	100.00	99.03	97.33	97.59	97.27
4	100.00	100.00	99.78	99.85	99.72	99.91	99.91	99.72
5		99.77	99.73	98.13		96.73	95.73	87.27
6		100.00	100.00	100.00		79.16	74.71	70.23
7		99.86	99.97	100.00		96.94	93.85	92.32
8		100.00	99.97	99.46		93.56	92.26	91.44
9			100.00	100.00			94.08	82.55
10			99.78	99.94			99.39	98.87
11			99.81	98.95			98.26	98.69
12			100.00	100.00			99.78	90.55
13				100.00				86.27
14				100.00				97.15
15				99.83				54.73
16				100.00				99.96
OA	99.98	99.96	99.95	99.74	99.79	97.07	97.08	96.44
AA	99.99	99.95	99.92	99.76	99.64	95.13	95.35	90.28
Kappa	99.97	99.95	99.94	99.71	99.66	96.15	96.56	95.82
Forgetting	0.00	−0.01	0.02	0.25	0.00	0.98	1.55	4.19

4.1.2. Training settings

We employed Adam optimizer with a learning rate of 0.001 and weight decay of 0.005. Training used a batch size of 256 and a memory size of 480, with per-class memory allocated proportionally to training samples. Following [31], we used the herding approach for sample selection. The initial stage required 200 epochs for all datasets except SA (50 epochs), while incremental stages including the integration and compression processes used 200 epochs each. Our CNN architecture follows [32], comprising four convolutional layers and three pooling layers.

All experiments are operated on the Ubutnu 20.04 system. The GPU used in training is Nvidia RTX 4090. The model is built with pytorch 1.11.0, and torchvision 0.12.0.

4.2. Evaluation results

We evaluated FEICA-CIL on four datasets with random seed 1. Tables 3–4 present classification results and forgetting rates across incremental stages. Performance metrics include overall accuracy (OA), average accuracy (AA), and Kappa coefficient, along with per-class accuracies. To quantify catastrophic forgetting, we calculated the forgetting rate as the difference between historical best and current performance for each class. A positive forgetting rate indicates knowledge loss in previously learned classes, while a negative rate signifies an improved performance in these classes.

On the SA dataset, our method maintained high OA (99.98%–99.74%) across tasks 1–4, with maximum degradation and increasing forgetting rate in the final task. Despite rapid forgetting rate increased on the Hanchuan dataset, the model achieved 96.44% final OA, showing only a 3.35% decrease across all tasks. The Longkou dataset exhibited gradual OA decline with increasing forgetting rates, yet limited the final performance drop to 0.37% with a maximum forgetting rate of 1.23%. Similarly, the PU dataset showed a mere 0.76% accuracy decrease with a maximum forgetting rate of 1.43%. Overall, FEICA-CIL effectively balanced knowledge retention and acquisition, achieving state-of-the-art results despite persistent catastrophic forgetting. Additionally, the four datasets encompassed diverse land covers and environmental conditions. Experimental results across these varied scenarios proved the effectiveness of our method under different environmental factors.

4.3. Comparative experiments

Methods marked with * share our memory management strategy, and we included GS_2FIN-CIL, the current state-of-the-art method for hyperspectral data classification. We conducted experiments with three random seeds (1, 2048, and 3407) and reported means and standard deviations of OAs, adopting the comparative criterion of reporting statistical results as in [33]. Tables 5–6 present detailed results with the best performances highlighted. On the SA dataset, FEICA-CIL consistently outperformed all competitors across all tasks, achieving a notable performance advantage of 0.54% in task 4 compared to the second-best method (Bic* at 98.97%). For the PU dataset, our approach demonstrated superior performance with accuracies of 100.00%, 99.71%, and 99.10% across tasks, surpassing the best competitor FORSTER*. Experiments on the Longkou dataset further confirmed our method's effectiveness, with FEICA-CIL achieving a 0.33% accuracy improvement in the final task over Bic*, which achieved the second highest performance. Notably, on the Hanchuan dataset, FEICA-CIL consistently demonstrated superior performance across all tasks, with the most substantial margin of 1.65% in the final task compared to Bic*, which ranked as the second most effective approach.

Across all datasets, our method not only achieved higher mean accuracies but also maintained competitive standard deviations comparable to or better than competing approaches, demonstrating both superior performance and robust stability across different random initializations. This consistency in performance underscores the reliability of our proposed approach in diverse experimental settings. It should be noted that results for GS_2FIN-CIL were directly obtained from the original paper as no source code was publicly available, and therefore means and standard deviations from multiple experimental runs are not available for this method. For all other methods, we conducted three independent experiments with different random seeds to ensure fair and robust comparison. Figs. 4–5 visualize classification maps comparing BiC, PODNet, FORSTER, DRC, and FEICA-CIL, demonstrating our method's consistent superiority across CIL tasks.

We compared computational costs for both training and inference across methods, as shown in Table 7. For inference, we evaluated model size (parameters) and computation complexity (FLOPs). While most methods maintain identical core architectures, our method and PODNet use a local similarity classifier, increasing model size to 182.25 KB compared to 164.25 KB in other methods. However, this only marginally increased computational requirements to 24.1 MFLOPs, just 0.2 MFLOPs above other methods. Regarding training costs, while our

Table 4
Classification results of FEICA-CIL on the Longkou and PU datasets (%).

Dataset	Longkou			PU		
Class	Task1	Task2	Task3	Task 1	Task 2	Task 3
1	99.87	99.59	99.85	100.00	99.84	99.94
2	100.00	100.00	100.00	100.00	99.99	99.96
3	96.98	96.82	90.98	100.00	99.95	97.69
4		99.79	99.75		98.05	92.67
5		98.47	97.07		100.00	99.23
6		99.72	99.81		100.00	99.98
7			99.99			99.92
8			98.87			99.83
9			87.81			97.93
OA	99.71	99.63	99.34	100.00	99.80	99.24
AA	98.95	99.07	97.13	100.00	99.64	98.57
Kappa	99.26	99.43	99.13	100.00	99.71	99.00
Forgetting	0.00	0.15	1.23	0.00	0.07	1.43

Table 5
Comparative results of different methods on the SA and PU datasets (%).

Dataset	SA				PU		
Method	Task 1	Task 2	Task 3	Task 4	Task 1	Task 2	Task 3
iCaRL [31]	99.98 ± 0.01	84.30 ± 4.08	67.44 ± 2.70	60.97 ± 4.14	99.94 ± 0.03	89.86 ± 3.34	85.46 ± 4.69
iCaRL* [31]	99.98 ± 0.01	82.61 ± 3.45	63.18 ± 2.15	57.29 ± 5.46	99.94 ± 0.03	89.66 ± 0.44	83.09 ± 0.62
Bic [34]	99.98 ± 0.02	99.85 ± 0.12	99.16 ± 1.15	89.79 ± 8.37	99.94 ± 0.05	91.00 ± 7.62	89.81 ± 4.54
Bic* [34]	99.98 ± 0.02	99.90 ± 0.06	99.79 ± 0.12	98.97 ± 0.24	99.94 ± 0.05	99.53 ± 0.08	98.77 ± 0.28
PODNet [22]	99.97 ± 0.03	93.38 ± 3.59	92.86 ± 1.27	84.81 ± 1.69	99.99 ± 0.01	80.00 ± 1.74	72.92 ± 1.12
PODNet* [22]	99.97 ± 0.03	98.34 ± 0.39	96.87 ± 1.03	92.31 ± 1.19	99.99 ± 0.01	94.65 ± 1.91	88.83 ± 4.19
FORSTER [23]	99.91 ± 0.15	99.83 ± 0.06	99.58 ± 0.09	96.92 ± 0.36	99.79 ± 0.03	98.90 ± 0.27	98.30 ± 0.11
FORSTER* [23]	99.91 ± 0.15	99.74 ± 0.04	99.35 ± 0.14	94.86 ± 3.42	99.79 ± 0.03	99.63 ± 0.03	99.09 ± 0.17
DRC [35]	99.97 ± 0.04	96.15 ± 5.56	92.56 ± 5.52	93.89 ± 1.09	99.84 ± 0.06	94.54 ± 0.82	97.62 ± 0.89
DRC* [35]	99.97 ± 0.04	95.61 ± 6.53	94.69 ± 7.86	89.78 ± 7.56	99.84 ± 0.06	98.00 ± 0.92	97.87 ± 0.42
GS_2FIN-CIL [26]	98.80	98.16	98.25	96.66	98.82	90.04	90.06
Ours	**99.99** ± **0.01**	**99.97** ± **0.01**	**99.95** ± **0.01**	**99.51** ± **0.23**	**100.00** ± **0.01**	**99.71** ± **0.08**	**99.10** ± **0.12**

Table 6
Comparative results of different methods on the Longkou and Hanchuan datasets (%).

Dataset	Longkou			Hanchuan			
Method	Task 1	Task 2	Task 3	Task 1	Task 2	Task 3	Task 4
iCaRL [31]	99.71 ± 0.05	82.18 ± 2.18	80.71 ± 6.45	99.55 ± 0.12	89.63 ± 1.42	66.79 ± 1.67	72.08 ± 0.81
iCaRL* [31]	99.71 ± 0.05	82.56 ± 0.75	70.86 ± 2.11	99.55 ± 0.12	89.08 ± 1.39	77.87 ± 0.86	71.20 ± 2.11
Bic [34]	99.63 ± 0.07	99.32 ± 0.16	97.78 ± 0.36	99.51 ± 0.07	97.08 ± 0.15	95.32 ± 0.36	92.70 ± 0.64
Bic* [34]	99.63 ± 0.07	99.40 ± 0.05	98.94 ± 0.06	99.61 ± 0.11	96.97 ± 0.16	96.24 ± 0.28	94.46 ± 0.23
PODNet [22]	99.70 ± 0.05	82.38 ± 4.56	86.05 ± 1.95	99.66 ± 0.05	87.93 ± 1.06	80.43 ± 2.84	72.21 ± 1.48
PODNet* [22]	99.70 ± 0.05	90.97 ± 1.89	92.45 ± 1.47	99.66 ± 0.05	93.14 ± 0.39	87.17 ± 2.04	83.18 ± 2.78
FORSTER [23]	99.57 ± 0.09	99.02 ± 0.20	98.40 ± 0.27	99.49 ± 0.08	88.80 ± 1.03	92.91 ± 0.72	93.37 ± 0.05
FORSTER* [23]	99.57 ± 0.09	98.91 ± 0.72	89.40 ± 8.25	99.49 ± 0.08	96.02 ± 0.17	64.52 ± 7.44	32.23 ± 1.01
DRC [35]	99.74 ± 0.10	98.99 ± 0.08	95.84 ± 2.31	99.57 ± 0.10	89.44 ± 0.22	87.26 ± 2.02	83.28 ± 2.49
DRC* [35]	99.74 ± 0.10	99.13 ± 0.10	95.95 ± 2.42	99.57 ± 0.10	96.16 ± 0.49	92.58 ± 2.69	85.66 ± 0.81
Ours	**99.79** ± **0.07**	**99.67** ± **0.04**	**99.27** ± **0.12**	**99.75** ± **0.04**	**97.32** ± **0.27**	**97.13** ± **0.06**	**96.11** ± **0.32**

method required higher GPU memory due to feature space expansion and integration, it achieved faster training times compared to similar methods with growing architectures, such as FORSTER and DRC. Although iCaRL, BiC, and PODNet showed quicker optimization, our method delivered superior classification accuracy. For a fair comparison, all methods were implemented with the same memory management strategy as FEICA-CIL. The results are based on experiments on the SA dataset. The FLOPs value represents the number of floating-point operations performed by the model after completing the final task.

4.4. Ablation study

This section presents a comprehensive analysis of several key components of our method. We first examined the impact of SSMA and investigated how varying memory sizes affect performance. We then compared against a full-precision network to assess accuracy degradation caused by quantization. Additionally, we evaluated the effectiveness of trainable parameters in feature integration. Finally, we analyzed the impact of correlation difference distillation by comparing models trained with and without this strategy.

4.4.1. Impacts of SSMA

To evaluate the impact of SSMA, we conducted experiments retaining only CentreCropResize and HorizontalFlip techniques in data preprocessing, while removing the SSMA strategy. Table 8 presents the Overall Accuracies (OAs) for all tasks across all datasets. Notable performance degradation was observed in the final task across all datasets, with the SA dataset showing the most significant decline of 1.62%. These results demonstrated that the combination of SSMA with other data augmentation techniques contributes significantly to the effectiveness of the proposed FEICA-CIL.

4.4.2. Impact of memory size

The effectiveness of our IL approach depends on the number of samples retained from previous tasks, as these samples are crucial for maintaining the classification capabilities on the old data. To evaluate

Table 7
Illustration of the training and inference consumption of different methods.

Method	Param (KB)	FLOPs (M)	Training time (s)				Memory consumption (MB)			
			Task1	Task2	Task3	Task4	Task1	Task2	Task3	Task4
iCaRL*	164.25	23.9	2	13	14	13	2232	2530	2534	2534
Bic*	164.28	23.9	2	11	13	12	2216	2550	2554	2554
PODNet*	182.25	24.1	2	16	17	16	2238	2551	2549	2589
FORSTER*	164.25	23.9	18	153	154	156	2228	2520	2538	2566
DRC*	164.25	23.9	18	157	158	157	2234	2518	2590	2614
FEICA-CIL	182.25	24.1	13	84	83	82	2707	2921	2931	2941

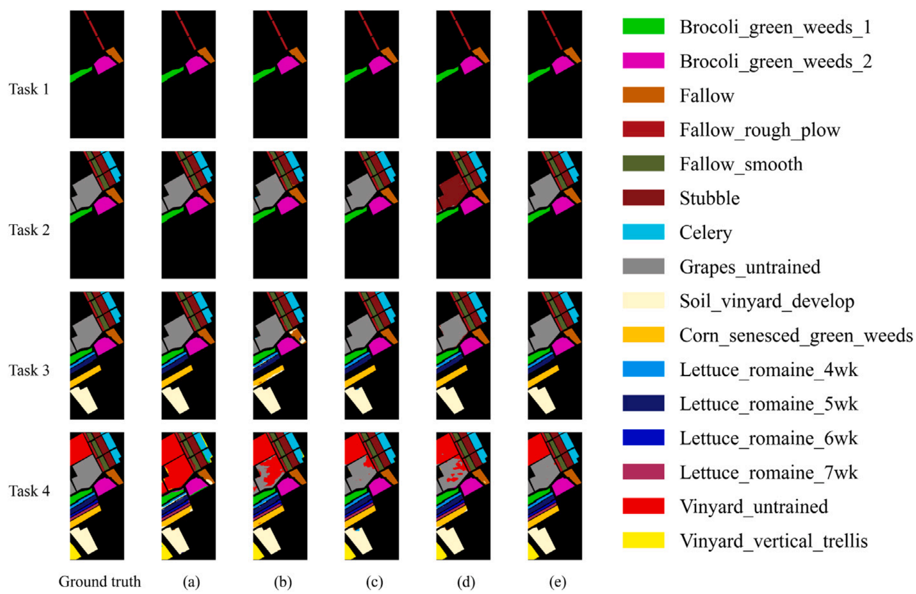

Fig. 4. Classification maps of comparative methods on the SA dataset. (a) Bic, (b) PODNet, (c) FORSTER, (d) DRC, (e) FEICA-CIL.

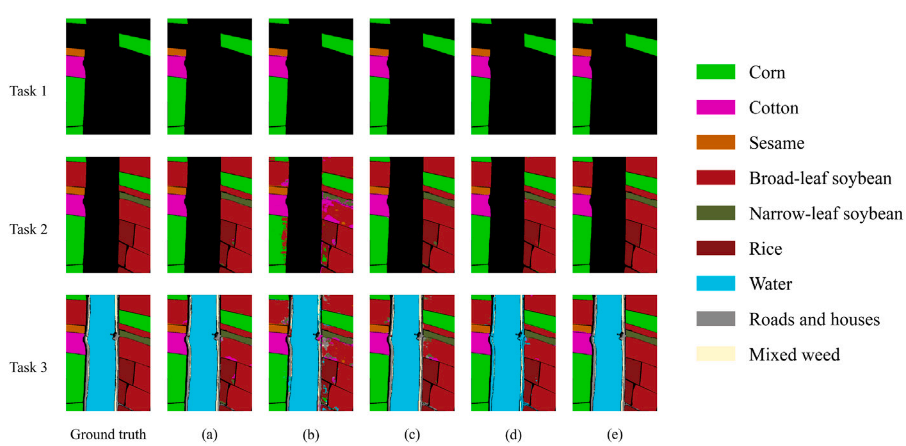

Fig. 5. Classification maps of comparative methods on the Longkou dataset. (a) Bic, (b) PODNet, (c) FORSTER, (d) DRC, (e) FEICA-CIL.

this dependency, we conducted extensive experiments with varying memory sizes across multiple datasets. The results presented in Table 9 reveal that increasing the memory size generally benefited model performance in most cases, while limited memory conditions typically led to lower classification accuracy. Notably, even under extreme memory constraints, our method demonstrated robust resistance to catastrophic forgetting, maintaining acceptable classification performance throughout the incremental learning process. These results validate the framework's resilience and practical applicability across diverse memory requirements.

4.4.3. Impacts of the quantization strategy

We adopted Deep Projection (DP) [36] as our quantization strategy, which employs projection layers and normalization functions to generate flexible gradients for quantized weights. To evaluate the effectiveness of quantization, we conducted a comparative analysis

Table 8
Comparison of the performance between FEICA-CIL trained with or without SSMA (%).

Dataset	SSMA	Task1	Task2	Task3	Task4
SA	with	99.98	99.96	99.95	99.74
	w/o	100.00	99.94	99.94	98.12
PU	with	100.00	99.80	99.24	
	w/o	99.98	99.66	99.19	
Longkou	with	99.71	99.63	99.34	
	w/o	99.69	99.54	98.85	
Hanchuan	with	99.79	97.07	97.08	96.44
	w/o	99.72	97.03	96.84	96.23

Table 9
Comparison of the performance between FEICA-CIL trained with different memory sizes (%).

Memory Size	120	240	480	960
SA	91.07	97.20	99.74	99.57
PU	94.91	97.58	99.24	99.55
Longkou	97.58	98.86	99.34	99.28
Hanchuan	90.07	93.74	96.44	97.17

Table 10
Comparison of the performance between FEICA-CIL structures based on the quantized and full precision networks (%).

Dataset	Network	Param (KB)	Task 1	Task 2	Task 3	Task 4
SA	Quantized	182.25	99.98	99.96	99.95	99.74
	Full precision	729.00	100.00	99.83	99.88	98.14
PU	Full precision	173.50	100.00	99.80	99.24	
	Full precision	694.00	100.00	99.55	98.53	
Longkou	Quantized	173.50	99.71	99.63	99.34	
	Full precision	694.00	99.82	99.71	99.20	
Hanchuan	Quantized	182.25	99.79	97.07	97.08	96.44
	Full precision	729.00	99.83	97.72	97.27	96.42

between the quantized and full-precision implementations of FEICA-CIL, with the distillation function removed in the initial stage for the latter. Table 10 presents the experimental results. The quantized model achieved comparable or superior performance while maintaining a reduced parameter count. Notably, it outperformed the full-precision model in the final task across four datasets, with the most substantial improvement of 1.60% observed on the SA dataset. The parameter counts for both full-precision and quantized models after the final task are documented in the table.

4.4.4. Impacts of the trainable factors

We examined the influence of trainable factors η and μ, which control the relative contributions of previous and new models in the final feature representations. Through experiments with varying scale coefficients (where zero coefficients represent direct output accumulation), we analyzed their impact. Table 11 presents our results, with default parameters in bold. The findings reveal that FEICA-CIL demonstrated robust performance across various combinations of η and μ, while omitting these factors results in notable performance degradation. These trainable parameters enhanced incremental learning outcomes by enabling dynamic weighting of previous and new model outputs.

4.4.5. Effect of the correlation difference distillation strategy

The proposed correlation difference distillation strategy aims to enhance the performance of the quantized model using an online teacher model. Experiments on the complete datasets are conducted to compare the results with and without the proposed strategy. As shown in Table 12, the distillation approach benefited the precision of the model across all datasets, with the largest improvement of 0.68% on the SA dataset.

Table 11
Comparison of FEICA-CIL with different settings of trainable factors (%).

Dataset	μ				η			
	0	0.1	1	2	0	0.1	1	2
SA	99.17	99.07	**99.74**	99.72	98.59	98.79	**99.74**	98.95
PU	99.10	99.32	**99.24**	98.81	99.04	98.87	**99.24**	99.18
Longkou	99.23	99.15	**99.34**	99.14	99.26	99.17	**99.34**	99.28
Hanchuan	96.07	96.51	**96.44**	96.31	96.05	95.93	**96.44**	95.99

Table 12
Comparison of the model's performance with and without the correlation distillation strategy (%).

Method	SA	PU	Longkou	Hanchuan
with	99.67	99.66	99.48	97.06
w/o	98.99	99.64	99.46	96.99

5. Conclusion

In this study, a novel CIL framework based on hyperspectral data named FEICA-CIL has been developed. To address the challenges posed by changing land covers and sequential data, the proposed method expands the feature space by supplementing a new model to learn new classes. The old knowledge is preserved by freezing the old model and integrating its outputs with the new model. A compression strategy is introduced to transfer all acquired knowledge to a single-branch objective model to mitigate the growing model parameters. Furthermore, a spatial–spectral mixed augmentation technique is introduced to help the models to learn more representative characteristics from spatiospectral information. While our approach adapts structural modification principles and knowledge distillation concepts from conventional CIL methods, it incorporates tailored designs specifically for HSIC challenges, resulting in state-of-the-art performance across various datasets. In future work, we plan to explore further methods to decrease the number of new network parameters. We aim to leverage the abundant parameters from the old model to simplify the IL stage.

CRediT authorship contribution statement

Ran Wu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Huanyu Liu:** Supervision, Project administration. **Zongcheng Yue:** Visualization, Data curation. **Chiu-Wing Sham:** Validation, Supervision. **Jun-Bao Li:** Supervision, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Claude in order to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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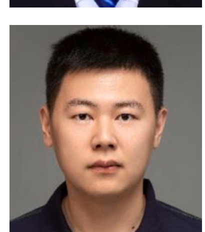

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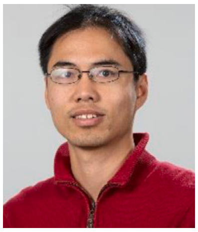

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